

Research Article

Applying a Hybrid Gray Wolf-Enhanced Whale Optimization Algorithm to the Capacitated Vehicle Routing Problem

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The study presents a novel hybrid gray wolf and whale optimization algorithm (hGWOAM) for the capacitated vehicle routing problem (CVRP). By integrating the enhanced whale optimization algorithm (EWOA) and gray wolf optimizer (GWO) with tournament selection, opposition-based learning, and mutation techniques, hGWOAM enhances routing efficiency under capacity constraints. Computational evaluations demonstrate its superior performance, achieving lower percentage deviations (% dev) compared to existing algorithms across multiple case studies and real-world applications. In Case Study 1, hGWOAM achieved a mean percentage deviation (%dev) lower than EWOA (0.89%), GWO (0.74%), SCA (0.59%), DA (1.63%), ALO (2.26%), MHPSO (1.85%), PSO (1.96%), DPGA (2.85%), and SGA (4.14%). In Case Study 2, hGWOAM outperformed EWOA (12.05%), GWO (2.53%), ALO (21.07%), and DA (17.58%). In a real-world application, it achieved the best %dev, surpassing EWOA (6.64%), GWO (6.34%), ALO (9.01%), and DA (12.24%). These findings highlight hGWOAM's potential for optimizing logistics, reducing operational costs, and minimizing environmental impact while also paving the way for future advancements in metaheuristic optimization.

Keywords: CVRP; decision-making support; enhanced whale optimization algorithm (hGWOAM); hybrid gray wolf

1. Introduction

In recent years, metaheuristic algorithms have emerged as powerful tools for solving complex optimization problems, particularly in technical and engineering domains. Many of these algorithms are inspired by animal behaviors and natural phenomena, offering innovative solutions to challenging problems. For example, Pham and Nguyen Dang [1] explored the Portia spider algorithm, a novel swarm-based technique derived from the sophisticated problem-solving behaviors of Portia spiders. Similarly, the effectiveness of the recently proposed graylag goose optimization algorithm was demonstrated by El-Kenawy and Khodadadi [2] in solving various optimization tasks. Zhao et al. [3] introduced the sea-horse optimizer, drawing inspiration from sea horses'

movements, predation strategies, and breeding behaviors, thereby highlighting its potential as an effective swarm intelligence-based metaheuristic.

The application of metaheuristic algorithms to complex problems has increasingly attracted significant research attention [4–7]. For example, two particle swarm optimization (PSO)-based algorithms explicitly tailored for unmanned aerial vehicle (UAV) networks were introduced by Arafat and Moh [8]. The first, known as the 3-D SIL algorithm, enhanced localization accuracy and reduced convergence time. The second algorithm, the energy-efficient SIC algorithm, optimized intercluster and intracluster distances, residual energy, and geographic positioning, thereby significantly improving network performance. In subsequent research, Arafat and Moh [9] proposed bio-inspired inclusion (BIL) and bio-

inspired clustering (BIC) localization schemes designed for forest fire detection and monitoring, both inspired by the hybrid gray wolf optimizer (HGWO). Additionally, a joint routing and charging strategy (JRCS) was developed in recent studies [10] to extend drone flight times by incorporating charging stations, effectively optimizing routes through clustering, segmentation, and mixed-integer linear programming (MILP). Furthermore, Lu et al. [10] analyzed urban low-carbon passenger transportation using the VPOSR model, employing game theory and CRITIC-entropy methods for indicator weighting, and applying the technique for order preference by similarity to ideal solution (TOPSIS) method to evaluate data collected from Tianjin between 2017 and 2021.

Beyond nature-inspired algorithms, several metaheuristic methods have emerged based on mathematical and physical concepts [11–14], facilitating a balanced approach between exploitation and exploration to efficiently address practical optimization problems. Hybrid algorithms have also received substantial attention for their ability to enhance the capabilities of traditional metaheuristics, with promising results reported across numerous engineering benchmarks and real-world applications [15–17]. Among various optimization challenges, the vehicle routing problem (VRP) is widely recognized as a particularly complex issue in logistics and transportation management, drawing extensive research interest. It encompasses multiple variants, including capacity constraints [18–22], time windows [23, 24], cost minimization [25], and management of delivery and pickup operations [26–28] for both homogeneous and heterogeneous fleets [29, 30]. A diverse range of approaches has been proposed to tackle these variants, such as MILP [31], dynamic programming [32], genetic algorithms (GA) [18, 25, 33], ant colony systems (ACS) [34], hybrid sine cosine algorithms (HSCA) [21, 35], hybrid particle swarm optimization (HPSO) [36], and the modified salp swarm algorithm (mSSA) [37].

Despite these advances, the existing research predominantly focuses on static VRP scenarios involving predetermined constraints [37–41], lacking adaptability to the dynamic and unpredictable nature of contemporary logistics environments. Furthermore, current algorithms frequently target specific VRP variants effectively [42–44] but struggle to seamlessly integrate multiple constraints, particularly when sustainability and fuel efficiency considerations are involved [45]. The evolving demands of global supply chains emphasize the urgent need for robust and flexible optimization algorithms capable of solving the CVRP under diverse scenarios while ensuring global optimization. Consequently, the development of advanced algorithms capable of addressing multiobjective and dynamic VRP challenges remains a critical area for future research. To address these challenges, this study introduces the hybrid gray wolf and enhanced whale optimization algorithm (hGWOAM), integrating the strengths of the enhanced whale optimization algorithm (EWOA) [46] and the gray wolf optimizer (GWO) [47], enhanced by the tournament selection (TS) strategy [48], opposition-based learning (OBL) [48], and mutation techniques [49]. Although both

whale optimization algorithm (WOA) and GWO have demonstrated success in various optimization contexts, their direct application to CVRP has revealed limitations, particularly regarding balancing exploration–exploitation dynamics and effectively managing the complex constraints typical in logistics operations.

By combining the distinctive features of these algorithms, hGWOAM aims to overcome such limitations, promoting a more adaptive, efficient, and sustainable logistics management framework. Specifically, global exploration and local exploitation are effectively balanced, improving both solution quality and convergence speed. The incorporation of OBL expands the search space by evaluating opposite solutions, while mutation strategies introduce controlled randomness to preserve population diversity and prevent premature convergence. Furthermore, the TS strategy increases selection pressure, ensuring the propagation of high-quality solutions across generations. Given the increasing complexity and environmental considerations in global supply chains, the development of advanced optimization tools capable of navigating modern logistics systems is essential. By integrating EWOA and GWO within the hGWOAM framework, this study presents a novel and effective solution to CVRP, overcoming the shortcomings of existing algorithms and addressing the dual challenges of sustainability and operational efficiency.

The proposed hGWOAM undergoes rigorous evaluation through computational experiments and real-world case studies, including scenarios with various capacity constraints. Its performance is compared to state-of-the-art algorithms, demonstrating superior accuracy, convergence speed, and robustness in solving complex, multiobjective optimization problems. This research advances the field of metaheuristic optimization and offers valuable insights for improving logistics and supply chain management. The findings pave the way for further developments in multiobjective and dynamic VRP optimization, highlighting hGWOAM's potential impact in both academic and industrial contexts. In the following sections, the nature and mathematical formulation of CVRP are presented in detail. Section 3 discusses computational experiments and results. Section 4 summarizes key findings, while Section 5 concludes with a discussion of the limitations and future research directions.

2. Model Development

2.1. Overview of CVRP. Operations research and logistics extensively recognize the importance of the CVRP [50], which involves efficiently distributing goods from a centralized warehouse to a defined set of clients while ensuring that vehicles return to their starting point. A formal description of this problem, as provided by Pham and Nguyen [21], includes several critical assumptions and constraints (Eq. (0)–Eq. (3)), which are detailed in Table 1.

2.1.1. Assumptions

- **Stable Demand:** The demand at each client's location is constant and predictable, facilitating accurate routing and scheduling.

TABLE 1: Mathematical formulations and constraints for CVRP.

Mathematical formulation	No.	Description variable	Formula meaning
$\text{Min } L = \sum_{i=0}^k \sum_{j=0}^k \sum_{t=1}^h d_{ij} z_{ijt}$	(0)	L : total distance d_{ij} represents the distance between locations i and j	Objective function
$\sum_{i=0}^k z_{ijt} = w_{jt} = w_{it}; \forall j = 1, 2, \dots, k; \forall t = 1, 2, \dots, h$	(1)	z_{ijt} is a binary variable indicating whether the route from i to j used by vehicle t w_{jt}, w_{it} are binary variables indicating whether customer j or i is served by vehicle t k is the total number of customers	Visit consistency: Each customer i, j is visited exactly once by exactly one vehicle
$\sum_{i=0}^k m_i w_{it} \leq m_t w_{it}; \forall t = 1, 2, \dots, h$	(2)	h denotes the total number of vehicles m_t represents the goods demanded by customer i	Capacity constraints: The total goods transported by each vehicle t does not exceed its capacity
$\sum_{i=1}^h w_{it} = \begin{cases} 1, \forall i = 1, 2, 3, \dots, k \\ 0, \forall i = h; \end{cases}$	(3)	p_i represents the capacity of vehicle t	Vehicle assignment: Each vehicle starts and ends its route at the warehouse, and exactly h vehicles are used

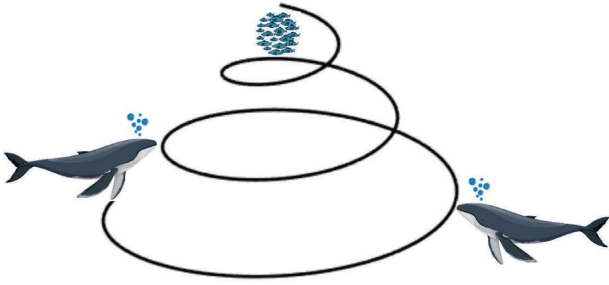


FIGURE 1: Bubble-net feeding technique used by humpback whales.

- **Fixed Vehicle Capacity:** Vehicles have a specified maximum capacity that must not be surpassed to prevent overloading and promote efficient and safe transportation.
- **Single Service Visit:** Each client receives a single visit from the fleet, maximizing efficiency in both time and resource use.

2.1.2. Constraints

- **Capacity Limits:** No vehicle route can transport a load exceeding the vehicle's maximum capacity.
- **Exclusive Customer Delivery:** Each customer receives delivery only once, ensuring precise and organized distribution.

2.1.3. Objective

- The primary goal is to reduce the total travel distance across all vehicles while adhering to the delivery requirements.

2.2. Algorithm Development for CVRP

2.2.1. EWOA. In 2016, Mirjalili and Lewis [51] introduced the WOA, an optimization method inspired by the bubble-net feeding behavior of humpback whales, where they swim in spirals to trap prey (see Figure 1). The WOA models this behavior using three main mechanisms:

- **Encircling Prey Strategy:** Simulated whales encircle prey, narrowing the search area and guiding candidate solutions toward optimal regions (Eq. (4) to Eq. (7)).
- **Spiral Bubble-Net Attacking Strategy:** This combines encircling with spiral movements to tighten the bubble net, enhancing exploitation near the best solution (Eq. (8) to Eq. (10)).
- **Search for Prey Strategy:** Whales randomly explore the search space, increasing solution diversity and preventing premature convergence (Eq. (11) to Eq. (12)).

The WOA is effective but faces challenges such as premature convergence, low population diversity, and an imbalance in search strategies. To overcome these issues, the EWOA was developed, with its mathematical formulation

detailed in Table 2. The EWOA introduces a pooling mechanism and three advanced search strategies:

- **Pooling Mechanism:** At each iteration, a Pool matrix is created at 1.5 times the size of the position matrix, enhancing diversity through crossover. For example, with 100 whales, the Pool matrix contains 150 whales. Each position P_i^t in the Pool is determined by blending the best X_{best}^t and the worst X_{worst}^t positions using binary vectors B_i^t and $\bar{B}_i^t$ (Eq. (13)).
- **Migrating Search Strategy:** This strategy randomly moves 20% of the whale population to unexplored regions to enhance exploration (Eq. (14)). It uses random positions X_{rand}^t and positions near the best solution X_{best}^t , increasing population diversity.
- **Preferential Selecting:** This approach updates positions using Eq. (15), influenced by dynamic coefficients A_i^t and C_i^t with P_{rand1}^t and P_{rand2}^t randomly chosen from the Pool to promote exploration.
- **Enriched Encircling Prey Strategy:** This strategy improves exploitation and convergence by adjusting positions with Eq. (16) to Eq. (18). It modifies the distance D^t to the best solution, incorporating a random Pool element to prevent stagnation.

These enhancements enable the EWOA to balance exploration and exploitation, increasing its effectiveness in solving complex optimization problems and enhancing the likelihood of finding global optima.

2.2.2. GWO. Mirjalili et al. [47] introduced the GWO in 2014, inspired by the social hierarchy and hunting strategies of wild wolves. The algorithm is structured into four ranks: alpha, beta, delta, and omega, with the top three representing the best solutions and omega ensuring population diversity (see Figure 2). GWO simulates two main wolf behaviors: encircling and hunting.

- **Encircling Period:** Wolves surround their prey by adjusting their positions based on the prey's location and specific coefficient vectors, guiding them to encircle the target and limit its escape (Eq. (19) to Eq. (22) in Table 3).
- **Hunting Phase:** Alpha, beta, and delta wolves lead the pack toward the prey using their experience to estimate its location, influencing the movement of other wolves (Eq. (23) to Eq. (25) in Table 3).

During the encircling and hunting phase, a random vector $\vec{a}$ is generated within $[-2a, 2a]$. If $|\vec{a}| < 1$, wolves attack a randomly chosen prey, initiating the "mining stage". If $|\vec{a}| > 1$, they abandon their current target to explore better options.

2.2.3. TS, OBL, and Mutation TS. TS is a popular method used in various optimization algorithms, including GA [52], and ant lion optimizer (ALO) [53]. This method is preferred

TABLE 2: Mathematical formulations and variable descriptions for EWOA.

Mathematical formulation	No.	Description variable
$\vec{D} = \vec{C} \times \vec{X}^*(t) - \vec{X}(t) $	(4)	$\vec{D}$: distance vector between the prey and the whale
$\vec{X}(t+1) = \vec{X}^*(t) - \vec{A} \times \vec{D}$	(5)	$\vec{X}(t)$: position vector of the current search agent. t current iteration
$\vec{A} = 2a \times r - a$	(6)	$\vec{X}^*(t)$: position vector of the best solution found so far
$\vec{C} = 2 \times r$	(7)	$\vec{A}$: coefficient vector, updated to reduce the difference between the best solution and current positions
$\vec{X}(t+1) = \vec{D} \times e^{bl} \times \cos(2\pi l) + \vec{X}^*(t)$	(8)	$\vec{C}$: coefficient vector that helps in the exploration and exploitation process
$\vec{D} = \vec{X}(t) - \vec{X}(t) $	(9)	$\vec{a}$: linearly decreases from 2 to 0 during iterations, used in calculating $\vec{A}$
$\vec{X}(t+1) = \begin{cases} \vec{X}^*(t) - \vec{A} \times \vec{D} & \text{if } p < 0.5 \\ \vec{D} \times e^{bl} \times \cos(2\pi l) + \vec{X}^*(t) & \text{if } p \geq 0.5 \end{cases}$	(10)	$\vec{r}$: random vector with elements in the range $[0, 1]$
$\vec{D} = \vec{C} \times \vec{X}_{rand} - \vec{X} $	(11)	$\vec{X}_{rand}$: random position vector from the existing population of whales
$\vec{X}(t+1) = \vec{X}_{rand} - \vec{A} \times \vec{D}$	(12)	X_i^{t+1} : position of the whale in the next iteration
$P_i^t = B_i^t \times X_{brand}^t + \vec{B}_i^t \times X_{worst}^t$	(13)	D^t : modified distance used in the enriched encircling prey strategy
$X_{i+1}^t = X_i^t - X_{brand}^t$	(14)	P_i^t : position of each whale in the Pool matrix, determined based on the best and worst positions
$X_{i+1}^t = X_i^t + A_i^t \times (C_i^t \times P_{rand1}^t - P_{rand2}^t)$	(15)	B_i^t : binary vector used in the pooling mechanism
$C_i^t = 2 \times rand$	(16)	$\vec{B}_i^t$: inverse binary vector of B_i^t
$X_{i+1}^t = X_{best}^t - A_i^t \times D^t$	(17)	X_{brand}^t : position of the best random whale in the Pool matrix
$D^t = C_i^t \times X_{best}^t - P_{rand3}^t $	(18)	X_{worst}^t : position of the worst whale in the Pool matrix

$P_{rand1}^t, P_{rand2}^t, P_{rand3}^t$ are randomly chosen positions from the Pool matrix
 C_i^t : preferential selecting strategy
 A_i^t : calculated using cauchy distribution for preferential selecting strategy.
 p : probability used to switch between the contraction-encircling mechanism and the spiral model
 b : constant used in the spiral updating position, impacting the shape of the logarithmic spiral
 l : random number in the interval $[-1, 1]$, used in the spiral updating position equation

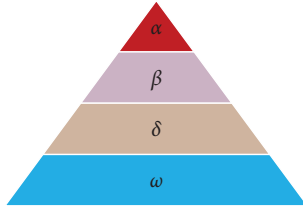


FIGURE 2: Hierarchy chart of a gray wolf population.

due to its simplicity and efficiency. In TS, k individuals are randomly selected from the population to compete against each other. The individual with the highest fitness among them is chosen and added to the next generation. The tournament size k , which is the number of individuals participating in each tournament, is a critical parameter. However, increasing the tournament size is known to lead to a higher expected loss of diversity [54]. In this study, a tournament size of $k=3$ is chosen.

Figure 3 illustrates the TS process, which preserves diversity and controls the convergence rate by giving all individuals equal selection opportunities. TS offers fast execution, especially in parallel implementations, and reduces the risk of domination by a few individuals. It also eliminates the need for fitness scaling or sorting, increasing efficiency. In hGWOAM, TS selects the best solution from k candidates in the solution Pool. Using Eq. (26) and Eq. (27) in Table 4, the fitness of all solutions is then evolved based on parameters from the top-performing solution. This process promotes exploration and maintains population diversity.

2.2.3.1. OBL. OBL is illustrated in Figure 4, where the opposite point x^* is determined by reflecting each coordinate of the initial point x across the midpoint between its lower and upper bounds. During the optimization process, both the original point x and its corresponding opposite point x^* are evaluated using the fitness function. The superior solution is then retained, while the less optimal one is discarded, as detailed in Eq. (28) in Table 4.

2.2.3.2. Mutation. Mutation plays a crucial role in introducing small random variations into offspring, thereby promoting population diversity and preventing premature convergence to local optima. During the mutation process, a mutation vector m_j^i is generated by selecting random components from the directional elements of the original vector x_i^i , as shown in Eq. (29) in Table 4. The values within the relevant domain of these directions are subsequently redistributed, as illustrated in Figure 5. The probability factor p_c is employed to maintain population diversity and mitigate the risk of premature convergence to local optima.

2.2.4. Algorithm Development. The CVRP is a discrete optimization challenge focused on optimizing customer visit sequences. The hGWOAM addresses this by integrating the GWO and the EWOA, utilizing strategies tailored for discrete problems.

Hybrid Strategy and Adaptations:

- **Integration of GWO and EWOA:** This combines the social hierarchy of gray wolves and the encircling behavior of humpback whales, enhancing exploration and exploitation of the search space for customer visit sequences.
- **TS:** This selects the best solutions through competitive comparison, increasing robustness.
- **OBL:** It considers both solutions and their opposites to accelerate convergence and explore unvisited areas.
- **Mutation Techniques:** Its custom mutations maintain genetic diversity, preventing premature convergence.

Procedure Overview:

- **Initialization:** A population of feasible solutions is generated representing sequences of customer visits per vehicle, evaluated by total distance and capacity constraints.
- **Leader Selection:** TS is used to identify effective solutions as leaders, guiding the search process.
- **Convergence and Adaptation:** OBL refines search directions by exploring both current and opposite solutions.
- **Iteration and Refinement:** GWO and EWOA behaviors with mutation are combined to explore new routes. The process continues until a stopping criterion (e.g., maximum iterations or solution stability) is reached.
- **Implementation Details:** Each iteration involves evaluating fitness, updating leaders through TS, enhancing search efficiency with OBL, and maintaining diversity with mutations. The algorithm dynamically adapts its strategy based on the current solution performance.
- **Termination:** The procedure ends when stopping criteria are met, yielding an optimal or near-optimal sequence of customer visits that minimizes total travel distance while satisfying constraints.

Figure 6 illustrates the schematic of the proposed hGWOAM method. The next section will apply this algorithm to solve the CVRP.

3. Computational Experiments

For the traveling salesman problem, the computational complexity grows exponentially with an increase in the number of cities. To clarify, a TSP involving n cities requires consideration of $1/2 * (n-1)!$ possible routes. The delivery problem is very complex, with $n=40$ which can have $4.07 * 10^{47}$ solutions. Such a large number of route permutations make the TSP particularly demanding in terms of computation. When examining the VRP, which fundamentally consists of multiple interconnected TSPs, the computational complexity increases significantly.

Table 5 presents the key parameter settings used for the CVRP model, including the population size, number of runs, maximum iterations, and algorithm execution time.

TABLE 3: Mathematical formulations and variable descriptions for GWO.

Mathematical formulation	No.	Description variable
$\vec{d}^t = \vec{c}^t \times \vec{a}^t - \vec{a}^t$	(19)	$\vec{x}^t$: position of a wolf at iteration t
$\vec{x}^{(t+1)} = \vec{x}^t - \vec{a} \times \vec{d}$	(20)	$\vec{d}$: vector indicating the distance to the prey
$\vec{a} = 2l \times r_1$	(21)	$\vec{a}$: randomly generated vector within $[-2a, 2a]$ determining whether wolves attack or search for new prey
$\vec{c} = 2 \times r_2$	(22)	$\vec{c}$: coefficient vector affecting wolf movement, taking values within $[0, 2]$ r_1, r_2 are random vectors
$\vec{d}_\alpha = \vec{x} - \vec{x}_\alpha $	(23)	$\vec{d}_\alpha$: distance to the alpha wolf's position
		$\vec{d}_\beta$: distance to the beta wolf's position
		$\vec{d}_\delta$: distance to the delta wolf's position
$\vec{x}_1, \vec{x}_2, \vec{x}_3$ are updated	(24)	$\vec{x}_1, \vec{x}_2, \vec{x}_3$ are updated position based on alpha, beta, delta wolf's influence.
$\vec{x}^{(t+1)} = (\vec{x}_1 + \vec{x}_2 + \vec{x}_3) / 3$	(25)	$\vec{x}^{(t+1)}$: average position of wolves for next iteration

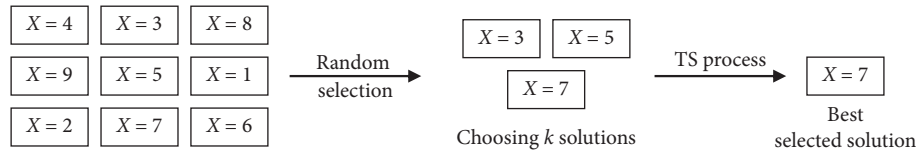


FIGURE 3: Illustration of the tournament selection method.

TABLE 4: Mathematical formulations of tournament selection, opposition-based learning, and the mutation method.

Mathematical formulation	No.	Description variable
Tournament selection:		
$x_i^g = \begin{cases} x_{Ts}^u & r_5 < C_{Ts} \\ x_i^u & r_5 \geq C_{Ts} \end{cases}$	(26)	x_i^u : the u^{th} parameter of solution i ; x_{Ts}^u : the u^{th} parameter of the solution chosen through TS process; C_{Ts} : the TS process condition
Subject to:		
$C_{Ts} = 0.1 \times (1 - i_{\text{current}}/I_{\text{max}})$	(27)	r_5 : a randomly generated value within $[0, 1]$
Opposition-based learning:		
$x^* = b_{\text{upper}} + b_{\text{lower}} - x$	(28)	b_{lower} : the lower boundary defining the range of x b_{upper} : the upper boundary defining the range of x
Mutation:		
$m_j^i = \begin{cases} \text{rand}(x_t^i \in [\text{Domain of } x_t^i]) & r_3 \leq p_c \\ x_t^i & r_3 > p_c \end{cases}$	(29)	$p_c = 0.1$ m_j^i is a mutation vector

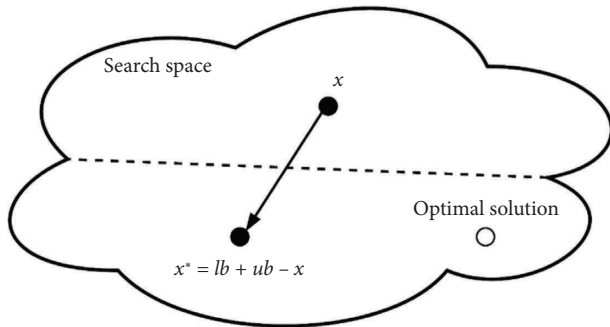


FIGURE 4: Illustration of the opposition-based learning method.

Population size refers to the number of candidate solutions generated in each iteration, while the number of runs ensures the reliability and consistency of results through multiple independent executions. The maximum number of iterations restricts the algorithm's search duration for an optimal solution, and execution time measures the total computational duration required to obtain the final solution.

Case Study 1: The hGWOAM algorithm completed its run in 23.38 s, slightly slower than EWOA (19.58 s) and GWO (20.80 s) but significantly faster than ALO (133.34 s) and DA (64.64 s). The relatively small problem size enabled all algorithms to find solutions quickly.

Case Study 2: With an increased population size of 100, 20 independent runs, and a maximum of 200 iterations, hGWOAM completed its execution in 393.88 s. This runtime was longer than those of EWOA (318.10 s) and GWO (369.52 s), but notably shorter compared to ALO (1772.28 s)

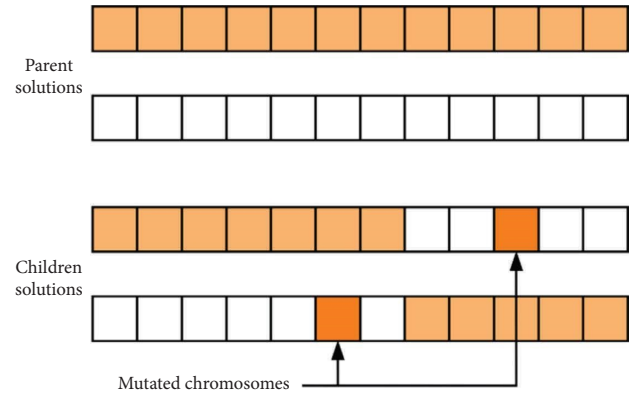


FIGURE 5: Illustration of the mutation method.

and DA (613.92 s). Although the EWOA was the fastest algorithm, hGWOAM delivered superior optimization results.

Case Study 3: In the most complex scenario, hGWOAM was executed in 1927.99 s, slightly longer than EWOA (1865.80 s), ALO (1738.56 s), and GWO (1873.26 s), though faster than DA (1937.52 s). Despite not being the fastest algorithm in this scenario, hGWOAM demonstrated competitive performance and achieved superior optimization results.

Overall, hGWOAM consistently demonstrated robust and efficient performance across all scenarios. Its effective balance between computational speed and solution quality makes it a reliable choice for solving CVRP problems. The next section will provide detailed performance analyses and solutions.

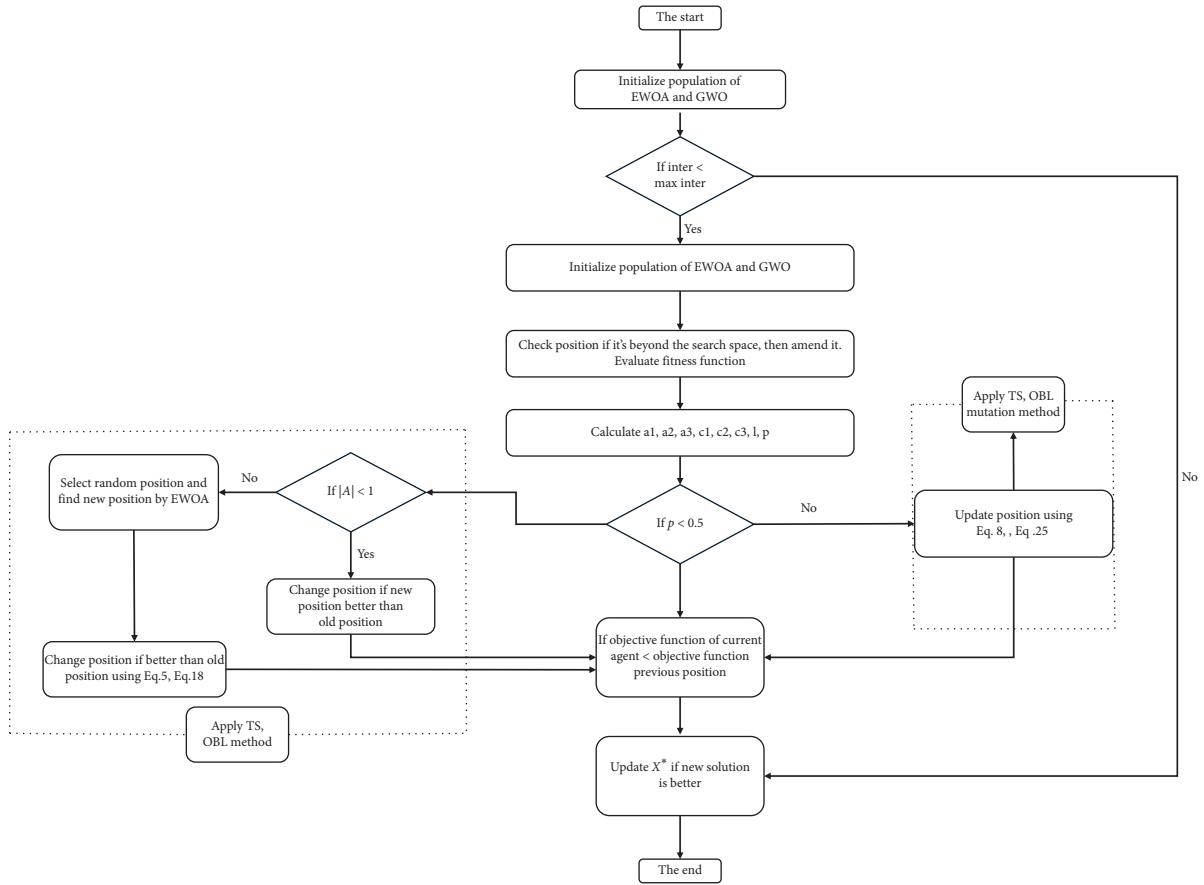


FIGURE 6: Flowchart of the proposed hGWOAM.

TABLE 5: Parameter settings for the CVRP problems.

No.	hGWOAM	EWOA	GWO	ALO	DA
Population size	20	20	20	20	20
Number of trials	20	20	20	20	20
Maximum number of iterations	50	50	50	50	50
Algorithm running time (s)	23.38	19.58	20.80	133.34	64.64
Population size	100	100	100	100	100
Number of trials	20	20	20	20	20
Maximum number of iterations	200	200	200	200	200
Algorithm running time (s)	393.88	318.10	369.52	1772.28	613.92
Population size	500	500	500	500	500
Number of trials	20	20	20	20	20
Maximum number of iterations	500	500	500	500	500
Algorithm running time (s)	1927.99	1865.80	1873.26	1738.56	1937.99

Note: The model runs on a Java platform with a 13th Gen Intel(R) Core(TM) i7-13700 2.10 GHz.

3.1. Case Study 1. This case study analyzes a distribution scenario where a central warehouse supplies eight customers using two delivery trucks, each with an 8-unit capacity. Table 6 presents distance data and customer demands. The goal is to optimize truck routes, minimizing total travel distance while adhering to all CVRP constraints.

In Table 7, the outcomes obtained through the application of various algorithms [21, 55], including EWOA, GWO, SCA, DA, ALO, MHPSO, PSO, DPGA, and SGA, are

presented. The hGWOAM demonstrates superior performance by achieving the lowest mean percentage deviation (%dev), indicating a more optimized solution. Specifically, the recorded %dev values for each algorithm are as follows: EWOA (0.89%), GWO (0.74%), SCA (0.59%), DA (1.63%), ALO (2.26%), MHPSO (1.85%), PSO (1.96%), DPGA (2.85%), and SGA (4.14%). These results clearly highlight the effectiveness of hGWOAM in achieving minimal deviation compared to other algorithms.

TABLE 6: Distance matrix and demand for Case Study 1 [55].

Node	0	1	2	3	4	5	6	7	8	Demand
0	0	4	6	7.5	9	20	10	16	8	
1	4	0	6.5	4	10	5	7.5	11	10	1
2	6	6.5	0	7.5	10	10	7.5	7.5	7.5	2
3	7.5	4	7.5	0	10	5	9	9	15	1
4	9	10	10	10	0	10	7.5	7.5	10	2
5	20	5	10	5	10	0	7	9	7.5	1
6	10	7.5	7.5	9	7.5	7	0	7	10	4
7	16	11	7.5	6	7.5	9	7	0	10	2
8	8	10	7.5	15	10	7.5	10	10	0	2

TABLE 7: Results of different algorithms in Case Study 1.

	Distribution of optimum solutions					Max	Min	Mean	%dev
hGWOAM	68	67.5	67.5	67.5	67.5	68	67.5	67.625	0.00
	67.5	67.5	67.5	67.5	67.5				
	67.5	68	67.5	68	68				
	68	67.5	67.5	67.5	67.5				
EWOA	69	67.5	68	67.5	68.5	70	67.5	68.225	0.89
	68.5	67.5	67.5	68.5	68.5				
	68.5	68	69	68	69.5				
	70	67.5	67.5	68	67.5				
GWO	68	67.5	68	67.5	68.5	69.5	67.5	68.125	0.74
	68.5	67.5	67.5	68.5	68.5				
	68.5	68	69	68	69.5				
	69	67.5	67.5	68	67.5				
SCA [21]	69	68	69	68	68	69.5	67.5	68.025	0.59
	68	69.5	67.5	67.5	68				
	69	68	67.5	67.5	67.5				
	68	67.5	68	67.5	67.5				
DA [21]	71.5	67.5	71.5	68	67.5	71.5	67.5	68.725	1.63
	69	70	70.5	68	69				
	70	67.5	67.5	69	68				
	67.5	68	68	67.5	69				
ALO [21]	71.5	68	71.5	68	67.5	71.5	67.5	69.150	2.26
	69	70	70.5	68	69				
	70	68	71	69	68				
	71.5	68	68	67.5	69				
MHPSO [55]	69.5	67.5	69	69	70	70	67.5	68.875	1.85
	69.5	70	69	67.5	67.5				
	69	69.5	69	70	67.5				
	70	69	67.5	70	67.5				
PSO [21]	67.5	70	70	69	69	70	67.5	68.950	1.96
	68	69	70	70	68.5				
	68.5	68.5	67.5	68	70				
	70	67.5	69.5	69	69.5				
DPGA [55]	70	69	67.5	71	69	72	67.5	69.550	2.85
	70.5	72	67.5	71.5	69				
	67.5	69	71	70	67.5				
	70.5	69	69.5	71	69				
SGA [55]	69	72	73.5	69	70	75.5	67.5	70.425	4.14
	71	67.5	69	69	75.5				
	70	69.5	69	73	69				
	74	70	69.5	69	70				

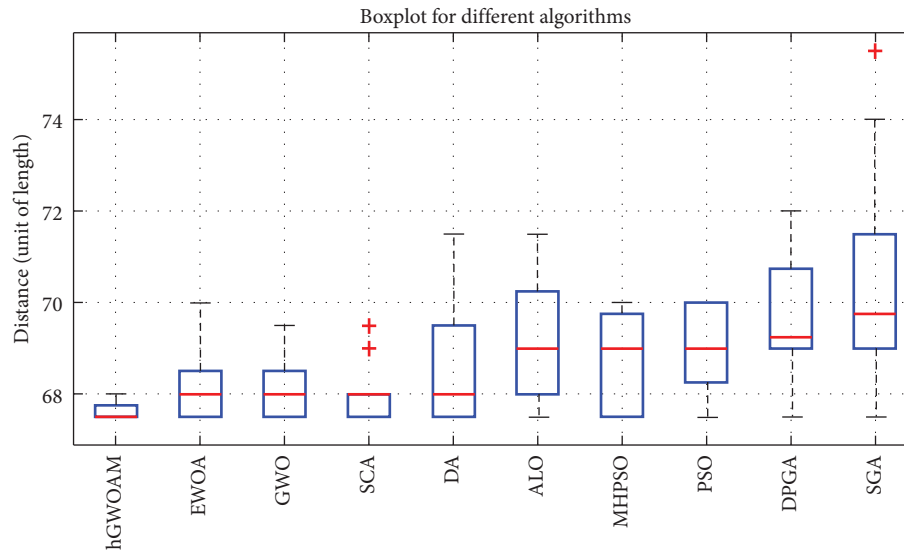


FIGURE 7: Boxplot of different algorithms in Case Study 1.

TABLE 8: Solution obtained by hGWOAM in Case Study 1.

Routes of the vehicles in Case Study 1		Distance
01	0 → Location 6 → Location 7 → Location 4 → 0	33.5
02	0 → Location 1 → Location 3 → Location 5 → Location 8 → Location 2 → 0	34

Note: Total distance: 67.5 units.

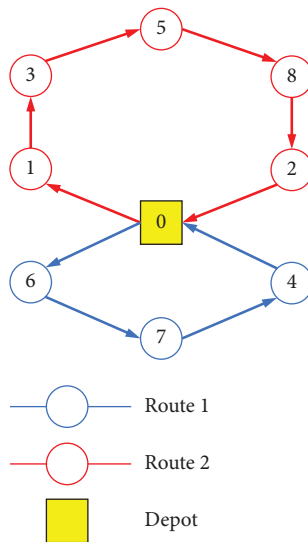


FIGURE 8: Best solution in Case Study 1.

While all algorithms yielded commendable results, hGWOAM demonstrated superior average performance and exhibited greater stability throughout both the exploration and exploitation phases. This conclusion is visually supported by Figure 7, which compares the data distribution of hGWOAM with those of other algorithms, clearly highlighting its advantages. Additionally, Table 8 presents the optimized navigation routes for the two vehicles as determined by hGWOAM, ensuring efficient delivery

operations. Figure 8 provides a detailed graphical representation of these routes, offering further insights into the algorithm's effectiveness in logistical optimization.

3.2. Case Study 2. In the second CVRP case study, a central warehouse supplies cement to 18 clients using three delivery trucks, each with a capacity of 500 cement bags. Table 9 provides detailed information regarding the distances between the warehouse and each client, as well as their specific delivery demands. The primary objective of this scenario is to optimize delivery routes by minimizing the total travel distance while strictly adhering to all CVRP constraints, including vehicle capacity and client delivery requirements. This case study further evaluates the effectiveness of advanced optimization algorithms in addressing realistic logistics challenges.

The results from different algorithm implementations are shown in Table 10. The hGWOAM's percentage deviation from the best solution is notably superior to that of other optimization techniques. Specifically, hGWOAM outperforms EWOA by 12.05%, GWO by 2.53%, ALO by 21.07%, and DA by 17.58%.

As shown in Figure 9, the data distribution obtained by the hGWOAM is compared with those of other algorithms, clearly revealing its superior performance. This figure illustrates how hGWOAM consistently outperforms competing algorithms across multiple trials. The results further highlight the effectiveness of hGWOAM in achieving an optimal solution, with a total travel distance of 414.39 units.

TABLE 9: Distance matrix and demand in Case Study 2.

Node	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Demand
0	0.0	36.1	33.5	28.3	25.0	31.6	53.9	36.1	20.6	16.8	42.7	32.7	27.6	53.0	50.0	30.4	58.9	47.6	48.3	0.0
1	36.1	0.0	60.2	10.0	60.2	67.1	20.0	0.0	35.0	25.3	11.2	33.0	50.0	25.1	14.1	61.8	59.9	50.5	23.4	100.0
2	33.5	60.2	0.0	50.2	38.1	25.0	80.2	60.2	25.5	35.0	70.0	66.2	10.8	65.3	71.6	10.0	91.8	80.7	78.6	100.0
3	28.3	10.0	50.2	0.0	53.2	58.3	30.0	10.0	25.0	15.5	20.6	34.5	40.0	26.2	22.4	52.2	63.2	52.8	31.8	95.0
4	25.0	60.2	38.1	53.2	0.0	18.0	76.3	60.2	41.2	41.2	65.2	45.5	39.3	77.9	74.3	29.2	62.6	53.0	67.8	65.0
5	31.6	67.1	25.0	58.3	18.0	0.0	85.4	67.1	39.1	43.6	74.3	60.1	30.7	79.9	80.6	15.0	80.1	69.9	79.4	80.0
6	53.9	20.0	80.2	30.0	76.3	85.4	0.0	20.0	55.0	45.2	11.2	38.6	70.0	33.3	14.1	81.4	58.2	51.7	15.1	100.0
7	36.1	0.0	60.2	10.0	60.2	67.1	20.0	0.0	35.0	25.3	11.2	33.0	50.0	25.1	14.1	61.8	59.9	50.5	23.4	140.0
8	20.6	35.0	25.5	25.0	41.2	39.1	55.0	35.0	0.0	10.8	45.3	48.1	15.0	41.4	46.1	29.2	76.6	65.3	55.1	70.0
9	16.8	25.3	35.0	15.5	41.2	43.6	45.2	25.3	10.8	0.0	35.0	38.3	25.2	37.0	37.7	36.7	67.2	56.0	44.4	50.0
10	42.7	11.2	70.0	20.6	65.2	74.3	11.2	11.2	45.3	35.0	0.0	29.7	60.1	32.3	15.0	70.7	53.2	45.1	12.8	80.0
11	32.7	33.0	66.2	34.5	45.5	60.1	38.6	33.0	48.1	38.3	29.7	0.0	59.4	58.0	44.1	62.6	29.2	18.4	25.5	40.0
12	27.6	50.0	10.8	40.0	39.3	30.7	70.0	50.0	15.0	25.2	60.1	59.4	0.0	54.6	61.0	17.2	86.5	75.2	69.4	50.0
13	53.0	25.1	65.3	26.2	77.9	79.9	33.3	25.1	41.4	37.0	32.3	58.0	54.6	0.0	19.2	70.5	84.7	75.5	44.7	50.0
14	50.0	14.1	71.6	22.4	74.3	80.6	14.1	14.1	46.1	37.7	15.0	44.1	61.0	19.2	0.0	74.3	68.2	60.1	26.2	50.0
15	30.4	61.8	10.0	52.2	29.2	15.0	81.4	61.8	29.2	36.7	70.7	62.6	17.2	70.5	74.3	0.0	86.5	75.6	78.0	80.0
16	58.9	59.9	91.8	63.2	62.6	80.1	58.2	59.9	76.6	67.2	53.2	29.2	86.5	84.7	68.2	86.5	0.0	11.3	43.1	70.0
17	47.6	50.5	80.7	52.8	53.0	69.9	51.7	50.5	65.3	56.0	45.1	18.4	75.2	75.5	60.1	75.6	11.3	0.0	36.7	120.0
18	48.3	23.4	78.6	31.8	67.8	79.4	15.1	23.4	55.1	44.4	12.8	25.5	69.4	44.7	26.2	78.0	43.1	36.7	0.0	60.0

TABLE 10: Results of different algorithms in Case Study 2.

Algorithm	Distribution of optimum solutions					Max	Min	Mean	Deviation ratio of the best solution (%)
hGWOAM	428.67	469.37	445.07	439.08	434.73	469.37	414.39	437.05	0.00
	442.62	434.73	419.84	416.92	427.55				
	439.85	421.27	448.65	452.30	414.39				
	458.90	428.32	428.64	445.07	444.93				
EWOA	565.55	591.92	621.43	611.39	581.64	625.29	464.33	560.21	12.05
	524.23	579.74	567.12	488.23	517.92				
	521.61	582.48	590.90	586.08	576.32				
	464.33	625.29	504.52	617.48	486.02				
GWO	485.32	453.27	429.34	433.00	472.71	491.84	424.89	462.70	2.53
	462.80	424.89	486.91	463.04	491.84				
	474.98	482.11	447.29	483.01	450.36				
	453.85	457.29	480.67	446.03	475.27				
ALO	555.00	572.14	556.54	528.74	523.04	572.92	501.71	537.31	21.07
	508.06	522.82	559.66	545.10	558.72				
	501.71	547.63	540.86	537.38	572.92				
	504.90	543.38	504.53	530.70	532.45				
DA	547.47	557.73	546.61	521.79	512.50	619.23	487.26	547.81	17.58
	525.57	551.76	522.92	578.71	578.71				
	574.23	619.23	541.45	556.40	551.08				
	529.87	537.71	600.11	515.10	487.26				

In Table 11, the delivery routes for the three trucks determined by hGWOAM are presented in detail. This table provides a comprehensive breakdown of the routes assigned to each truck, ensuring efficient delivery operations while strictly adhering to capacity constraints.

Figure 10 provides a graphical representation of these delivery routes. The practical application of the hGWOAM in optimizing the delivery process and minimizing travel distances is demonstrated visually by depicting the routes

taken by each truck. Overall, these results underscore the robustness of the hGWOAM in addressing complex logistical challenges, such as the CVRP.

3.3. Case Study 3. The third CVRP case study introduces increased complexity by expanding the problem to 200 customers, representing a realistic logistical challenge in cement distribution. In this scenario, a central warehouse supplies cement to these locations using 37 trucks, each with

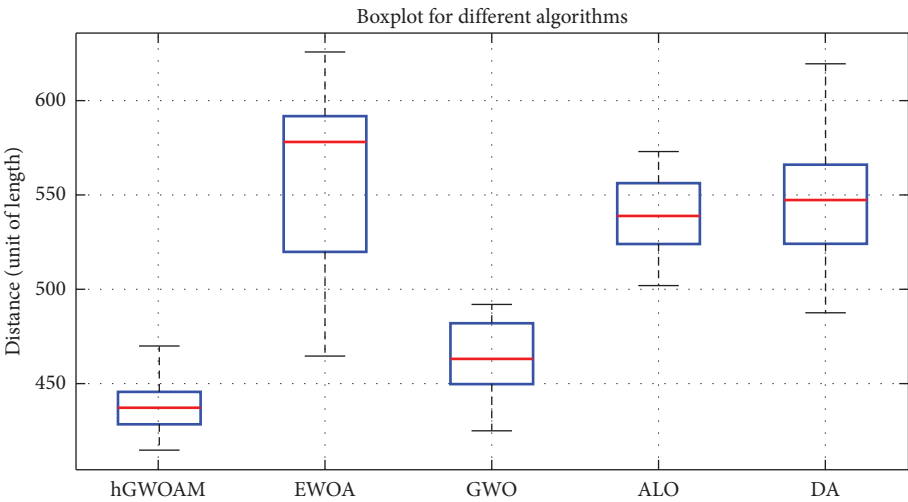


FIGURE 9: Boxplot of different algorithms in Case Study 2.

TABLE 11: Solution obtained by hGWOAM in Case Study 2.

Routes of the vehicles in Case Study 2		Distance
01	0 → Location 9 → Location → Location 13 → Location 14 → Location 7 → Location 3	125.41
02	0 → Location 11 → Location 17 → Location 16 → Location 18 → Location 6 → Location 10	174.53
03	0 → Location 8 → Location 12 → Location 2 → Location 15 → Location 5 → Location 4	114.45

Note: Total distance: 414.39 units.

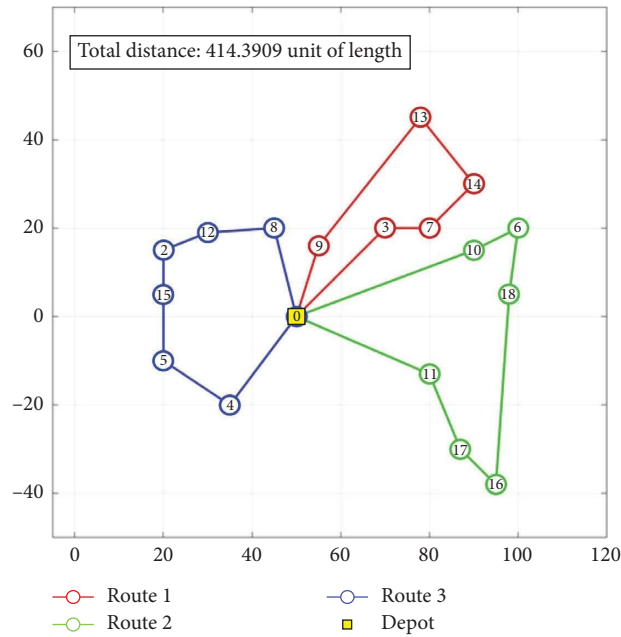


FIGURE 10: Vehicle routes obtained by hGWOAM in Case Study 2.

a capacity of 700 bags. The objective is to minimize the total delivery distance while ensuring that capacity constraints and timely deliveries are satisfied.

This scenario closely mirrors real-world logistics challenges, where efficient route planning contributes to cost savings, improved service levels, and enhanced operational

TABLE 12: Coordinates and demand in Case Study 3.

Location	Longitude	Latitude	Demand	Location	Longitude	Latitude	Demand
Deport	107.0242788	10.6072385		51	107.1637245	10.4986051	80
1	107.0837984	10.3451602	100	52	107.2048204	10.5213112	90
2	107.0904903	10.3609904	50	53	107.208381	10.5709611	100
3	107.0890006	10.3733413	80	54	107.2293859	10.5880569	100
4	107.0924625	10.3627624	90	55	107.2113585	10.592591	50
5	107.1021562	10.3754122	100	56	107.3080995	10.5853336	50
6	107.1031228	10.3761541	110	57	107.2205906	10.5753757	60
7	107.0913193	10.3617377	200	58	107.2034027	10.602037	60
8	107.1076677	10.378978	100	59	107.209525	10.5912688	80
9	107.089926	10.3598197	80	60	107.3120569	10.5829239	90
10	107.08311	10.3560193	90	61	107.2252112	10.643967	100
11	107.0396139	10.653406	100	62	107.2081765	10.5919773	70
12	107.1970048	10.5000715	80	63	107.1008667	10.3740802	80
13	107.0454262	10.6574723	60	64	107.1143352	10.3836713	100
14	107.1524361	10.5049466	70	65	107.126075	10.3901564	120
15	107.2710915	10.4903057	90	66	107.0871781	10.3561336	120
16	107.2205906	10.5753757	100	67	107.1539828	10.4214295	130
17	107.1691781	10.4940003	100	68	107.1466794	10.4103401	140
18	107.1679409	10.5209525	120	69	107.1240686	10.3892222	150
19	107.1710108	10.4992729	130	70	107.103371	10.3801157	180
20	107.1889633	10.4901892	100	71	107.139878	10.4089466	190
21	107.2095274	10.4082787	200	72	107.1673059	10.4170584	140
22	107.073832	10.4696444	180	73	107.1363939	10.3979967	150
23	107.2195898	10.4112071	90	74	107.1454289	10.3966029	120
24	107.2340333	10.3946979	100	75	107.0544285	10.6015609	110
25	107.2897264	10.4379579	120	76	107.0539763	10.5825626	90
26	107.2289028	10.4257346	150	77	107.0561958	10.5734413	130
27	107.0933073	10.6067698	150	78	107.0732812	10.539833	140
28	107.2376394	10.3975834	160	79	107.0860874	10.5172811	160
29	107.1976773	10.4857485	180	80	107.1693349	10.4757113	180
30	107.2230215	10.4159073	170	81	107.1512173	10.5008324	200
31	107.2765195	10.4990328	90	82	107.0760384	10.5339968	150
32	107.2425555	10.5379508	100	83	107.0908985	10.5210308	170
33	107.2710915	10.4903057	120	84	107.0707401	10.5889924	190
34	107.2712451	10.4902598	130	85	107.0561477	10.588926	210
35	107.2626432	10.4878332	70	86	107.0673998	10.5877707	200
36	107.2831723	10.4962767	50	87	107.0680263	10.5967365	160
37	107.2708947	10.4899472	80	88	107.055012	10.5971994	140
38	107.2653139	10.4880226	100	89	107.1889805	10.5097044	130
39	107.2617453	10.4889771	120	90	107.1968505	10.5280828	150
40	107.2665416	10.5410849	130	91	107.2454316	10.6987053	200
41	107.2301894	10.4684375	160	92	107.2478515	10.6777852	100
42	107.2315898	10.5288541	190	93	107.2483769	10.6726445	100
43	107.2901925	10.4988428	200	94	107.244858	10.6413062	100
44	107.1709489	10.495647	210	95	107.2632892	10.6497864	50
45	107.2312619	10.5284822	180	96	107.245854	10.6479095	80
46	107.2454199	10.4616401	170	97	107.3160785	10.64359	90
47	107.2897264	10.4379579	160	98	107.2426032	10.6354967	100
48	107.1789434	10.4968487	150	99	107.2451825	10.644182	110
49	107.1827772	10.4921791	120	100	107.2483382	10.6607925	200
50	107.1933209	10.4858346					
101	107.232978	10.64943	100	151	107.2945458	10.4267901	160
102	107.2403911	10.7038262	80	152	107.2930629	10.4295222	180
103	107.2419072	10.711214	90	153	107.3290068	10.4558186	170
104	107.2394116	10.7241377	100	154	107.285747	10.4441869	100
105	107.1546267	10.5797859	80	155	107.3116764	10.4894856	120
106	107.147047	10.5884566	60	156	107.2971875	10.5008125	120
107	107.1330498	10.590313	70	157	107.2144027	10.5402928	130
108	107.1250047	10.3780398	90	158	107.3832565	10.5583102	140
109	107.1911687	10.4079234	100	159	107.4316703	10.558723	150

TABLE 12: Continued.

Location	Longitude	Latitude	Demand	Location	Longitude	Latitude	Demand
110	107.1210467	10.4011201	100	160	107.3917763	10.5480988	180
111	107.0928753	10.4645698	120	161	107.3895491	10.5312328	190
112	107.0764286	10.4612866	130	162	107.2778042	10.4613389	140
113	107.2184636	10.4153714	100	163	107.0904814	10.5159272	150
114	107.2116025	10.4140367	200	164	107.090069	10.5209418	120
115	107.1250945	10.40374	180	165	107.0780857	10.5327865	110
116	107.3954402	10.5340722	90	166	107.1441511	10.4032053	90
117	107.4224733	10.5430976	100	167	107.1707011	10.5320209	130
118	107.5463775	10.5569513	120	168	107.1833661	10.6771288	140
119	107.423355	10.5655446	150	169	107.3983709	10.5261142	160
120	107.5527777	10.5793247	150	170	107.322347	10.4527448	180
121	107.5587113	10.5828861	160	171	107.2955838	10.4448747	200
122	107.3818004	10.6569732	180	172	107.3472876	10.5142798	150
123	107.3878668	10.6825681	170	173	107.4298217	10.4785718	170
124	107.5011562	10.6895802	100	174	107.4270886	10.4802238	190
125	107.4070582	10.7282229	50	175	107.2718451	10.5936879	50
126	107.0607605	10.5627161	80	176	107.2793802	10.5940728	80
127	107.0710316	10.546983	90	177	107.2958359	10.6747727	150
128	107.0485622	10.6328866	100	178	107.3150366	10.6431725	120
129	107.0548736	10.6049324	110	179	107.2569872	10.6461742	100
130	107.0595686	10.586266	200	180	107.2527158	10.6444889	50
131	107.1263476	10.6439379	100	181	107.253576	10.64427	80
132	107.0564189	10.6245086	80	182	107.242989	10.6457916	80
133	107.0924997	10.6410546	90	183	107.3278829	10.4528535	100
134	107.0696791	10.6305131	100	184	107.3275584	10.4529541	150
135	107.1046283	10.6437178	80	185	107.3272817	10.4587591	100
136	107.0846291	10.639186	60	186	107.4486567	10.5519968	120
137	107.1618843	10.6565236	70	187	107.4828032	10.5523471	180
138	107.1683386	10.6543573	90	188	107.4878438	10.5534228	200
139	107.331577	10.456957	100	189	107.5387238	10.5587742	100
140	107.4190301	10.5678459	100	190	107.5436277	10.5484656	50
141	107.3863474	10.5298422	120	191	107.4913489	10.5602509	60
142	107.2144733	10.4847797	130	192	107.4109591	10.5297061	80
143	107.2104375	10.4813125	100	193	107.4192092	10.5526763	90
144	107.2138066	10.4864488	200	194	107.1189525	10.4002784	100
145	107.2000512	10.4985376	180	195	107.1570879	10.4149267	60
146	107.3887808	10.5527956	90	196	107.2881782	10.498286	70
147	107.4390963	10.6099732	100	197	107.0422531	10.5957367	90
148	107.3740277	10.6372208	120	198	107.0531962	10.5771138	100
149	107.4491143	10.6291658	150	199	107.0716147	10.6252839	100
150	107.449656	10.6300841	150	200	107.0753104	10.6032123	100

efficiency. The increased number of customers and delivery vehicles highlights the necessity for advanced optimization algorithms such as hGWOAM to effectively handle complex logistical scenarios. Tables 12 provide detailed distance and delivery data relevant to this case study.

Table 13 presents the results obtained from the implementations of various algorithms. It can be observed that the best %dev solution achieved by the hGWOAM surpasses those obtained by other optimization techniques. Specifically, hGWOAM outperformed EWOA by 6.64%, GWO by 6.34%, ALO by 9.01%, and DA by 12.24%. These results demonstrate that the hGWOAM successfully achieved the optimal solution, with a total travel distance of 2863.96 units. This outcome confirms the algorithm's capability to effectively solve the CVRP by minimizing the overall travel distance while satisfying all associated constraints.

Figure 11 visually represents the data distribution, clearly illustrating the superior performance of the hGWOAM compared to other methods. The results indicate that hGWOAM consistently outperforms competing algorithms across multiple trials, highlighting its robustness and efficiency. Additionally, Table 14 provides a detailed breakdown of the delivery routes for the 37 trucks determined by hGWOAM. This detailed breakdown ensures efficient delivery operations while strictly adhering to vehicle capacity constraints, further validating the algorithm's effectiveness in optimizing complex logistics scenarios.

Figure 12 illustrates the delivery routes of 37 trucks, highlighting the practical application of the hGWOAM in optimizing deliveries and minimizing travel distances. In combination with the data presented in Table 14, this visualization underscores the algorithm's robustness and

TABLE 13: Results of different algorithms in Case Study 3.

Algorithm	Distribution of optimum solutions					Max	Min	Mean	Deviation ratio of the best solution (%)
hGWOAM	2863.96	2892.86	2914.98	3000.21	3030.52	3052.89	2863.96	2951.60	0.00
	2873.59	2900.75	2923.55	3007.57	3038.39				
	2882.05	2905.34	2925.46	3019.51	3046.98				
	2886.03	2908.91	2930.15	3028.23	3052.89				
EWOA	3054.13	3140.71	3194.87	3255.81	3301.45	3331.68	3054.13	3214.00	6.64
	3102.33	3157.54	3211.25	3265.80	3305.71				
	3117.27	3166.39	3231.79	3276.90	3319.91				
	3127.36	3179.11	3242.31	3297.63	3331.68				
GWO	3045.55	3106.41	3149.81	3187.03	3249.45	3298.97	3045.55	3168.40	6.34
	3060.49	3120.93	3157.10	3203.55	3270.55				
	3073.44	3124.80	3169.68	3214.54	3285.50				
	3093.16	3139.54	3176.67	3240.72	3298.97				
ALO	3121.94	3163.51	3189.62	3224.95	3242.06	3269.31	3121.94	3202.14	9.01
	3138.57	3169.83	3199.52	3232.10	3253.05				
	3148.23	3179.54	3211.03	3235.62	3265.87				
	3158.36	3184.45	3217.14	3238.20	3269.31				
DA	3214.41	3299.00	3340.30	3374.80	3440.37	3467.00	3214.41	3354.49	12.24
	3245.03	3307.37	3349.24	3398.25	3450.57				
	3259.32	3316.18	3355.31	3405.33	3456.30				
	3287.20	3332.29	3365.85	3425.62	3467.00				

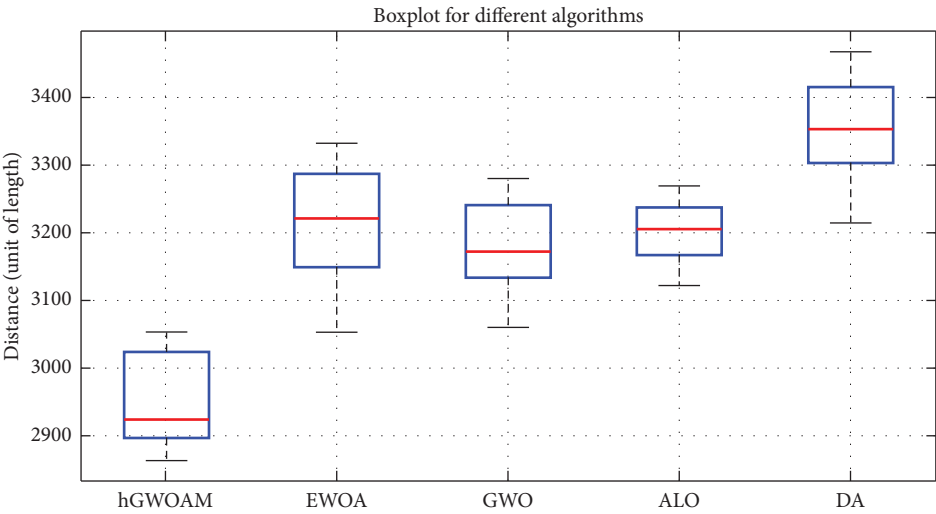


FIGURE 11: Boxplot of different algorithms in Case Study 3.

TABLE 14: Solution obtained by hGWOAM in Case Study 3.

Routes of the vehicles in Case Study 3		Distance
01	0 → Location 28 → Location 23 → Location 24 → Location 25 → Location 14 → Location 13 → Location 12 → 0	79.73
02	0 → Location 26 → Location 27 → Location 22 → Location 34 → 0	64.83
03	0 → Location 21 → Location 20 → Location 15 → Location 19 → Location 36 → Location 10 → 0	71.48
04	0 → Location 17 → Location 35 → Location 33 → Location 16 → Location 31 → Location 32 → Location 18 → 0	84.34
05	0 → Location 30 → Location 9 → Location 5 → Location 29 → Location 3 → Location 4 → 0	73.61

TABLE 14: Continued.

	Routes of the vehicles in Case Study 3	Distance
06	0 → Location 7 → Location 6 → Location 1 → Location 2 → Location 8 → Location 11 → 0	56.24
07	0 → Location 54 → Location 56 → Location 51 → Location 33 → Location 4 → 0	101.49
08	0 → Location 36 → Location 34 → Location 42 → Location 9 → Location 20 → 0	85.77
09	0 → Location 35 → Location 48 → Location 55 → Location 30 → 0	70.93
10	0 → Location 45 → Location 19 → Location 58 → Location 50 → Location 32 → 0	93.88
11	0 → Location 62 → Location 44 → Location 28 → Location 22 → Location 24 → Location 12 → 0	107.47
12	0 → Location 37 → Location 17 → Location 25 → Location 27 → Location 57 → Location 5 → 0	75.79
13	0 → Location 2 → Location 40 → Location 7 → Location 6 → 0	59.68
14	0 → Location 46 → Location 23 → Location 31 → Location 26 → Location 14 → Location 39 → 0	77.56
15	0 → Location 38 → Location 15 → Location 13 → Location 10 → Location 47 → 0	88.83
16	0 → Location 60 → Location 59 → Location 11 → Location 52 → Location 18 → Location 21 → Location 16 → 0	100.13
17	0 → Location 63 → Location 53 → Location 61 → Location 49 → Location 3 → 0	97.72
18	0 → Location 8 → Location 43 → Location 1 → Location 41 → Location 29 → 0	78.18
19	0 → Location 7 → Location 8 → Location 10 → Location 9 → Location 17 → Location 16 → Location 14 → 0	138.46
20	0 → Location 20 → Location 13 → Location 18 → Location 15 → 0	104.39
21	0 → Location 21 → Location 22 → Location 24 → Location 25 → Location 32 → 0	121.06
22	0 → Location 23 → Location 12 → Location 1 → Location 2 → Location 33 → 0	107.51
23	0 → Location 11 → Location 4 → Location 19 → Location 28 → Location 27 → Location 26 → 0	119.19
24	0 → Location 31 → Location 5 → Location 6 → Location 30 → Location 3 → Location 29 → 0	130.28
25	0 → Location 31 → Location 10 → Location 30 → Location 8 → Location 20 → 0	33.91
26	0 → Location 7 → Location 32 → Location 9 → Location 19 → Location 6 → Location 34 → 0	28.11
27	0 → Location 12 → Location 23 → Location 13 → Location 11 → 0	14.43
28	0 → Location 33 → Location 5 → Location 4 → Location 3 → Location 36 → Location 14 → Location 15 → 0	28.68
29	0 → Location 22 → Location 25 → Location 21 → Location 2 → Location 1 → 0	31.96
30	0 → Location 28 → Location 26 → Location 29 → Location 35 → 0	31.61
31	0 → Location 27 → Location 24 → Location 18 → Location 17 → Location 16 → 0	34.98
32	0 → Location 4 → Location 7 → Location 2 → Location 9 → Location 1 → 0	89.93
33	0 → Location 19 → Location 15 → Location 10 → Location 3 → Location 31 → 0	90.26
34	0 → Location 24 → Location 13 → Location 8 → Location 12 → Location 5 → Location 6 → Location 22 → 0	90.50
35	0 → Location 29 → Location 26 → Location 14 → Location 18 → Location 30 → Location 23 → Location 32 → 0	83.84
36	0 → Location 16 → Location 20 → Location 17 → Location 21 → Location 25 → Location 27 → 0	83.33
37	0 → Location 28 → Location 11 → 0	33.86

Note: Total distance 2863.96 (unit of length)

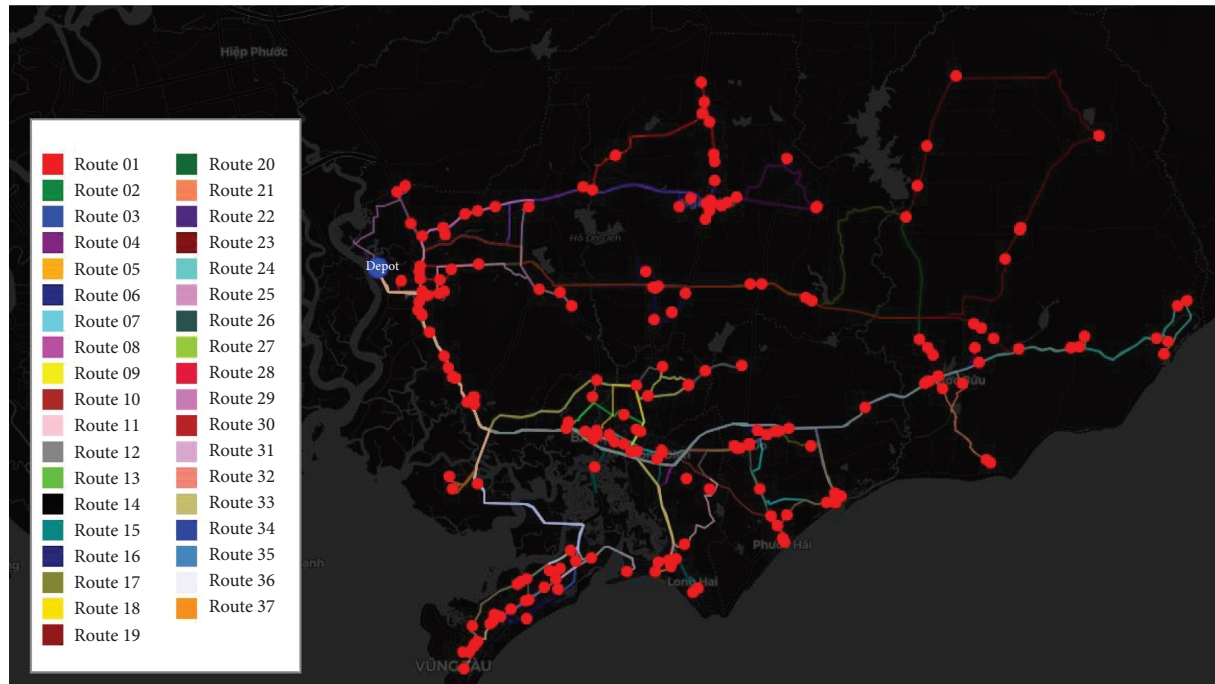


FIGURE 12: Vehicle routes obtained by hGWOAM in Case Study 3.

efficiency in addressing complex CVRP scenarios. The integration of detailed tabular results and graphical insights demonstrates how hGWOAM effectively manages multiple delivery routes, ensuring minimal total travel distance while adhering to vehicle capacity and delivery constraints. These results reinforce the algorithm's capability to efficiently handle real-world logistical challenges.

4. Conclusion

This research introduced hGWOAM, an innovative optimization framework combining the strengths of the EWOA and the GWO, further enhanced by TS, OBL, and mutation techniques. The proposed approach significantly improved solutions to the CVRP, effectively balancing global exploration and local exploitation to optimize logistical routes. When tested across three distinct CVRP scenarios, hGWOAM consistently outperformed existing algorithms, demonstrating robust adaptability to a variety of logistical challenges. These advanced features position hGWOAM as a valuable decision-support tool in logistics and construction management, substantially increasing operational efficiency and reducing costs. By enhancing route planning and adaptability, hGWOAM provides significant practical benefits for supply chain and logistics decision-making processes. The conducted case studies illustrated the algorithm's efficacy in addressing complex logistical and management scenarios:

- Case Study 1: Involving eight clients and two delivery vehicles, hGWOAM outperformed comparative algorithms, achieving the lowest mean percentage deviation (%dev). Specifically, it surpassed EWOA (0.89%), GWO (0.74%), SCA (0.59%), DA (1.63%),

ALO (2.26%), MHP SO (1.85%), PSO (1.96%), DPGA (2.85%), and SGA (4.14%).

- Case Study 2: With 18 customers and three delivery vehicles, hGWOAM produced the shortest delivery distance of 414.39 units, demonstrating substantial improvements in %dev compared to EWOA (12.05%), GWO (2.53%), ALO (21.07%), and DA (17.58%). This underscores its effectiveness in more complex logistic environments.
- Case Study 3: The most complex scenario involved delivering cement to 200 customers using 37 trucks. The hGWOAM achieved an optimal distance of 2863.96 units, outperforming EWOA by 6.64%, GWO by 6.34%, ALO by 9.01%, and DA by 12.24%. These results illustrate hGWOAM's scalability and high efficiency in addressing large-scale logistics problems.

The three case studies consistently confirmed the robustness and superior efficiency of hGWOAM in solving CVRP challenges, successfully minimizing total travel distances while fully meeting operational constraints. Overall, this research confirms hGWOAM's effectiveness as a powerful optimization method, providing significant contributions and implications for logistics and construction management.

5. Discussion of Limitations and Future Research Directions

Despite its strengths, the hGWOAM hybrid model has limitations, particularly its slow convergence in complex or high-dimensional optimization problems, which can lead to impractically long computation times. Future research could

address this issue by exploring adaptive parameter settings, dynamic population sizing, and hybridization with algorithms such as PSO or GA to improve the exploration–exploitation balance, enhance solution diversity, and increase computational efficiency. The algorithm's longer runtime and reduced performance in large-scale problems highlight the need for improved computational efficiency. Strategies such as code optimization, high-performance computing, heuristic methods, problem decomposition, and parallel processing could significantly enhance the scalability and effectiveness of hGWOAM. Implementing these improvements would broaden hGWOAM's applicability from specific calculations to a wide range of logistics and supply chain management tasks. By increasing versatility and robustness, these advancements would ensure that hGWOAM remains a valuable tool for logistics and construction management, capable of meeting complex industry demands.

Data Availability Statement

The corresponding author is available to provide the data, model, or code underlying the findings of this study upon request, in accordance with reasonable conditions.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

All authors, including Vu Hong Son Pham, Nghiep Trinh Nguyen Dang, and Van Nam Nguyen, jointly contributed to the writing of the main manuscript, preparation of all figures and tables, and reviewed and approved the final version prior to submission.

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References

- [1] V. H. S. Pham and N. T. Nguyen Dang, "Portia Spider Algorithm: An Evolutionary Computation Approach for Engineering Application," *Artificial Intelligence Review* 57, no. 2 (2024): 24–42, <https://doi.org/10.1007/s10462-023-10683-1>.
- [2] E.-S. M. El-Kenawy, N. Khodadadi, S. Mirjalili, A. A. Abdelhamid, M. M. Eid, and A. Ibrahim, "Greylag Goose Optimization: Nature-Inspired Optimization Algorithm," *Expert Systems With Applications* 238 (2024): 122147, <https://doi.org/10.1016/j.eswa.2023.122147>.
- [3] S. Zhao, T. Zhang, S. Ma, and M. Wang, "Sea-Horse Optimizer: A Novel Nature-Inspired Meta-Heuristic for Global Optimization Problems," *Applied Intelligence* 53, no. 10 (2023): 11833–11860, <https://doi.org/10.1007/s10489-022-03994-3>.
- [4] L. K. Singh, M. Khanna, S. Thawkar, and R. Singh, "Collaboration of Features Optimization Techniques for the Effective Diagnosis of Glaucoma in Retinal Fundus Images," *Advances in Engineering Software* 173 (2022): 103283, <https://doi.org/10.1016/j.advengsoft.2022.103283>.
- [5] L. K. Singh, M. Khanna, H. Garg, and R. Singh, "Efficient Feature Selection Based Novel Clinical Decision Support System for Glaucoma Prediction From Retinal Fundus Images," *Medical Engineering & Physics* 123 (2024): 104077, <https://doi.org/10.1016/j.medengphy.2023.104077>.
- [6] L. K. Singh, M. Khanna, H. Garg, and R. Singh, "Emperor Penguin Optimization Algorithm-and Bacterial Foraging Optimization Algorithm-Based Novel Feature Selection Approach for Glaucoma Classification From Fundus Images," *Soft Computing* 28, no. 3 (2024): 2431–2467, <https://doi.org/10.1007/s00500-023-08449-6>.
- [7] MunishKhanna, L. K. Singh, and H. Garg, "A Novel Approach for Human Diseases Prediction Using Nature Inspired Computing & Machine Learning Approach," *Multimedia Tools and Applications* 83, no. 6 (2024): 17773–17809.
- [8] M. Y. Arafat and S. Moh, "Localization and Clustering Based on Swarm Intelligence in UAV Networks for Emergency Communications," *IEEE Internet of Things Journal* 6, no. 5 (2019): 8958–8976, <https://doi.org/10.1109/jiot.2019.2925567>.
- [9] M. Y. Arafat and S. Moh, "Bio-Inspired Approaches for Energy-Efficient Localization and Clustering in UAV Networks for Monitoring Wildfires in Remote Areas," *IEEE Access* 9 (2021): 18649–18669, <https://doi.org/10.1109/access.2021.3053605>.
- [10] F. Lu, H. Hao, and H. Bi, "Evaluation on the Development of Urban Low-Carbon Passenger Transportation Structure in Tianjin," *Research in Transportation Business & Management* 55 (2024): 101142, <https://doi.org/10.1016/j.rtbm.2024.101142>.
- [11] L. Abualigah, A. Diabat, S. Mirjalili, M. Abd Elaziz, and A. H. Gandomi, "The Arithmetic Optimization Algorithm," *Computer Methods in Applied Mechanics and Engineering* 376 (2021): 113609, <https://doi.org/10.1016/j.cma.2020.113609>.
- [12] F. Rezaei, H. R. Safavi, M. Abd Elaziz, and S. Mirjalili, "GMO: Geometric Mean Optimizer for Solving Engineering Problems," *Soft Computing* 27, no. 15 (2023): 10571–10606, <https://doi.org/10.1007/s00500-023-08202-z>.
- [13] V. K. Patel and V. J. Savsani, "Heat Transfer Search (HTS): A Novel Optimization Algorithm," *Information Sciences* 324 (2015): 217–246, <https://doi.org/10.1016/j.ins.2015.06.044>.
- [14] S. Mirjalili, S. M. Mirjalili, and A. Hatamlou, "Multi-Verse Optimizer: A Nature-Inspired Algorithm for Global Optimization," *Neural Computing & Applications* 27, no. 2 (2016): 495–513, <https://doi.org/10.1007/s00521-015-1870-7>.
- [15] P. V. H. Son and N. D. N. Trinh, "Development of Novel Hybrid Multi-Verse Optimizer With Sine Cosine Algorithm for Better Global Optimization," *International Journal of Computational Intelligence and Applications* 23, no. 02 (2024): 2450002, <https://doi.org/10.1142/s1469026824500020>.
- [16] V. H. S. Pham, N. T. Nguyen Dang, and V. N. Nguyen, "Enhancing Engineering Optimization Using Hybrid Sine Cosine Algorithm With Roulette Wheel Selection and Opposition-Based Learning," *Scientific Reports* 14, no. 1 (2024): 694, <https://doi.org/10.1038/s41598-024-51343-w>.
- [17] S. Kumar, B. S. Yildiz, P. Mehta, et al., "Chaotic Marine Predators Algorithm for Global Optimization of Real-World

- Engineering Problems," *Knowledge-Based Systems* 261 (2023): 110192, <https://doi.org/10.1016/j.knsys.2022.110192>.
- [18] M. Akhand, Z. Jannat, and K. Murase, "Capacitated Vehicle Routing Problem Solving Using Adaptive Sweep and Velocity Tentative PSO," *International Journal of Advanced Computer Science and Applications* 8, no. 12 (2017): <https://doi.org/10.14569/ijacsa.2017.081237>.
 - [19] V. H. S. Pham, V. N. Nguyen, and N. T. N. Dang, "Innovative Hybrid Algorithm for Efficient Routing of Limited Capacity Vehicles," *Intelligent Systems With Applications* 25 (2025): 200491, <https://doi.org/10.1016/j.iswa.2025.200491>.
 - [20] V. H. S. Pham, V. N. Nguyen, and N. T. Nguyen Dang, "Hybrid Whale Optimization Algorithm for Enhanced Routing of Limited Capacity Vehicles in Supply Chain Management," *Scientific Reports* 14, no. 1 (2024): 793, <https://doi.org/10.1038/s41598-024-51359-2>.
 - [21] V. H. S. Pham and V. N. Nguyen, "Cement Transport Vehicle Routing With a Hybrid Sine Cosine Optimization Algorithm," *Advances in Civil Engineering* 2023, no. 1 (2023): 1–15, <https://doi.org/10.1155/2023/2728039>.
 - [22] V. H. S. Pham, N. T. N. Dang, and V. N. Nguyen, "Advanced Vehicle Routing in Cement Distribution: A Discrete Salp Swarm Algorithm Approach," *International Journal of Management Science and Engineering Management* 20, no. 1 (2025): 1–13, <https://doi.org/10.1080/17509653.2024.2324172>.
 - [23] M. F. Ibrahim, M. Mustofa, P. Meilanitasari, and S. Wijaya, "An Improved Ant Colony Optimization Algorithm for Vehicle Routing Problem With Time Windows," *Jurnal Teknik Industri* 23, no. 2 (2022): 105–120, <https://doi.org/10.22219/jtiumm.vol23.no2.105-120>.
 - [24] Ç. Koç, T. Bektaş, O. Jabali, and G. Laporte, "A Hybrid Evolutionary Algorithm for Heterogeneous Fleet Vehicle Routing Problems With Time Windows," *Computers & Operations Research* 64 (2015): 11–27, <https://doi.org/10.1016/j.cor.2015.05.004>.
 - [25] M. A. Mohammed, M. K. Abd Ghani, R. I. Hamed, S. A. Mostafa, M. S. Ahmad, and D. A. Ibrahim, "Solving Vehicle Routing Problem by Using Improved Genetic Algorithm for Optimal Solution," *Journal of Computational Science* 21 (2017): 255–262, <https://doi.org/10.1016/j.jocs.2017.04.003>.
 - [26] F. Lu, T. Yan, H. Bi, M. Feng, S. Wang, and M. Huang, "A Bilevel Whale Optimization Algorithm for Risk Management Scheduling of Information Technology Projects Considering Outsourcing," *Knowledge-Based Systems* 235 (2022): 107600, <https://doi.org/10.1016/j.knsys.2021.107600>.
 - [27] M. F. Ibrahim, M. Putri, D. Farista, and D. M. Utama, "An Improved Genetic Algorithm for Vehicle Routing Problem Pick-Up and Delivery With Time Windows," *Jurnal Teknik Industri* 22, no. 1 (2021): 1–17, <https://doi.org/10.22219/jtiumm.vol22.no1.1-17>.
 - [28] M. Huang, F.-Q. Lu, W.-K. Ching, and T. K. Siu, "A Distributed Decision Making Model for Risk Management of Virtual Enterprise," *Expert Systems With Applications* 38, no. 10 (2011): 13208–13215, <https://doi.org/10.1016/j.eswa.2011.04.135>.
 - [29] S.-H. Huang, Y.-H. Huang, C. A. Blazquez, and C.-Y. Chen, "Solving the Vehicle Routing Problem With Drone for Delivery Services Using an Ant Colony Optimization Algorithm," *Advanced Engineering Informatics* 51 (2022): 101536, <https://doi.org/10.1016/j.aei.2022.101536>.
 - [30] L. Wei, Z. Zhang, D. Zhang, and A. Lim, "A Variable Neighborhood Search for the Capacitated Vehicle Routing Problem With Two-Dimensional Loading Constraints," *European Journal of Operational Research* 243, no. 3 (2015): 798–814, <https://doi.org/10.1016/j.ejor.2014.12.048>.
 - [31] M. A. Rizal and I. Saidatuningtyas, "Model of Flexible Periodic Vehicle Routing Problem-Service Choice Considering Inventory Status," *Jurnal Teknik Industri* 22, no. 1 (2021): 125–137, <https://doi.org/10.22219/jtiumm.vol22.no1.125-137>.
 - [32] N. R. Sabar, A. Bhaskar, E. Chung, A. Turkey, and A. Song, "A Self-Adaptive Evolutionary Algorithm for Dynamic Vehicle Routing Problems With Traffic Congestion," *Swarm and Evolutionary Computation* 44 (2019): 1018–1027, <https://doi.org/10.1016/j.swevo.2018.10.015>.
 - [33] B. Rabbouch, F. Saâdaoui, and R. Mraihi, "Efficient Implementation of the Genetic Algorithm to Solve Rich Vehicle Routing Problems," *Operational Research* 21, no. 3 (2021): 1763–1791, <https://doi.org/10.1007/s12351-019-00521-0>.
 - [34] M. Reed, A. Yiannakou, and R. Evering, "An Ant Colony Algorithm for the Multi-Compartment Vehicle Routing Problem," *Applied Soft Computing* 15 (2014): 169–176, <https://doi.org/10.1016/j.asoc.2013.10.017>.
 - [35] V. H. S. Pham and V. N. Nguyen, "Cement Transport Vehicle Routing Problem With Hybrid Sine Cosine Optimization Algorithm in Construction Management," in *Proceedings of the Third International Conference on Sustainable Civil Engineering and Architecture* (Singapore: Springer Nature Singapore, 2024).
 - [36] M. Ali and B. N. I. Farida, "Completion of FCVRP Using Hybrid Particle Swarm Optimization Algorithm," *Jurnal Teknik Industri* 22, no. 1 (2021): 98–112, <https://doi.org/10.22219/jtiumm.vol22.no1.98-112>.
 - [37] V. H. Son Pham, N. T. Nguyen Dang, and V. N. Nguyen, "Advancing Large-Scale Cement Vehicle Distribution: The Modified Salp Swarm Algorithm," *International Journal of Systems Science: Operations & Logistics* 11, no. 1 (2024): 2305817, <https://doi.org/10.1080/23302674.2024.2305817>.
 - [38] D. M. Utama, B. Widjonarko, and D. S. Widodo, "A Novel Hybrid Jellyfish Algorithm for Minimizing Fuel Consumption Capacitated Vehicle Routing Problem," *Bulletin of Electrical Engineering and Informatics* 11, no. 3 (2022): 1272–1279, <https://doi.org/10.11591/eei.v11i3.3263>.
 - [39] D. M. Utama, B. N. I. Farida, U. Fitriani, M. F. Ibrahim, and D. S. Widodo, "Hybrid Henry Gas Solubility Optimization: An Effective Algorithm for Fuel Consumption Vehicle Routing Problem," *Jurnal Ilmiah Teknik Industri* 20, no. 2 (2021): 141–152, <https://doi.org/10.23917/jiti.v20i2.15640>.
 - [40] D. M. Utama, T. A. Fitria, and A. K. Garside, "Artificial Bee Colony Algorithm for Solving Green Vehicle Routing Problems With Time Windows," *Journal of Physics: Conference Series* 1933, no. 1 (2021): 012043, <https://doi.org/10.1088/1742-6596/1933/1/012043>.
 - [41] M. Ibrahim, F. Nurhakiki, D. Utama, and A. Rizaki, "Optimised Genetic Algorithm Crossover and Mutation Stage for Vehicle Routing Problem Pick-Up and Delivery With Time Windows," in *IOP Conference Series: Materials Science and Engineering* (IOP Publishing, 2021).
 - [42] D. M. Utama, W. O. N. Safitri, and A. K. Garside, "Modified Camel Algorithm for Optimizing Green Vehicle Routing Problem With Time Windows," *Jurnal Teknik Industri* 24, no. 1 (2022): 23–36, <https://doi.org/10.9744/jti.24.1.23-36>.
 - [43] D. M. Utama, U. Fitriani, I. Amallynda, and R. D. Azmi, "A Novel Hybrid Yellow Saddle Goatfish Algorithm for Fuel Consumption Vehicle Routing Problem With Simultaneous Pick-Up and Delivery Problem," *Jurnal Teknik Industri* 23, no. 1 (2022): 43–66, <https://doi.org/10.22219/jtiumm.vol23.no1.43-66>.

- [44] D. M. Utama, A. Yurifah, and A. K. Garside, "A Novel Hybrid Spotted Hyena Optimizer: An Algorithm for Fuel Consumption Capacitated Vehicle Routing Problem," *International Journal of Technology* 14, no. 5 (2023): 1049–1059, <https://doi.org/10.14716/ijtech.v14i5.5148>.
- [45] M. Y. Arafat and S. Moh, "JRCS: Joint Routing and Charging Strategy for Logistics Drones," *IEEE Internet of Things Journal* 9, no. 21 (2022): 21751–21764, <https://doi.org/10.1109/jiot.2022.3182750>.
- [46] M. H. Nadimi-Shahraki, H. Zamani, and S. Mirjalili, "Enhanced Whale Optimization Algorithm for Medical Feature Selection: A COVID-19 Case Study," *Computers in Biology and Medicine* 148 (2022): 105858, <https://doi.org/10.1016/j.compbimed.2022.105858>.
- [47] S. Mirjalili, S. M. Mirjalili, and A. Lewis, "Grey Wolf Optimizer," *Advances in Engineering Software* 69 (2014): 46–61, <https://doi.org/10.1016/j.advengsoft.2013.12.007>.
- [48] J. Tang and X. Zhao, "Particle Swarm Optimization With Adaptive Mutation," in *2009 WASE International Conference on Information Engineering* (IEEE, 2009).
- [49] F. Zhao, J. Zhang, J. Wang, and C. Zhang, "A Shuffled Complex Evolution Algorithm With Opposition-Based Learning for a Permutation Flow Shop Scheduling Problem," *International Journal of Computer Integrated Manufacturing* 28, no. 11 (2014): 1–16, <https://doi.org/10.1080/0951192x.2014.961965>.
- [50] C. Archetti, D. Feillet, M. Gendreau, and M. Grazia Speranza, "Complexity of the VRP and SDVRP," *Transportation Research Part C: Emerging Technologies* 19, no. 5 (2011): 741–750, <https://doi.org/10.1016/j.trc.2009.12.006>.
- [51] S. Mirjalili and A. Lewis, "The Whale Optimization Algorithm," *Advances in Engineering Software* 95 (2016): 51–67, <https://doi.org/10.1016/j.advengsoft.2016.01.008>.
- [52] D. E. Goldberg and K. Deb, "A Comparative Analysis of Selection Schemes Used in Genetic Algorithms," in *Foundations of Genetic Algorithms* (Elsevier, 1991), 69–93.
- [53] H. Kılıç and U. Yüzgeç, "Improved Antlion Optimization Algorithm via Tournament Selection and its Application to Parallel Machine Scheduling," *Computers & Industrial Engineering* 132 (2019): 166–186, <https://doi.org/10.1016/j.cie.2019.04.029>.
- [54] A. Sokolov and D. Whitley, "Unbiased Tournament Selection," in *Proceedings of the 7th Annual Conference on Genetic and Evolutionary Computation* (Washington, DC, 2005).
- [55] W. Zhengchu, L. Jun, Z. Muxun, and F. Jian, "Research in Capacitated Vehicle Routing Problem Based on Modified Hybrid Particle Swarm Optimization," in *2009 IEEE International Conference on Intelligent Computing and Intelligent Systems* (2009).